

NAME: _____ PERIOD: _____ DATE: _____

Significant Does Not Mean Caused

AP Statistics · Unit 3 · Topic 3.7 · B: 2026-12-07 · E: 2027-01-13

[STAR] TODAY'S PROBLEM

A cafeteria served 5,000 September trays after new sorting-bin labels were installed. The director claims at least 60% are sorted correctly. An SRS of 250 trays finds 136 correct; the club tests at $\alpha = 0.05$.

Imani: “The p -value is 0.0354, so reject H_0 . The labels caused sorting below 60 percent for every month.”

Drew: “The 5.6-point gap may be too small to matter, so it is not significant. Accept the director’s claim.”

September study

Source	Trays	Correct	α
September	5,000	claim: $\geq 60\%$	–
SRS	250	136	0.05

FIRST TAKE — before the video (5 min)

Which parts of Imani’s and Drew’s statements do you trust? Mark one statistical claim and one scope or importance claim to revisit.

[BOOK] MAKE THE DECISION, THEN BOUND THE CLAIM (keep on the page)

Compare the p -value with the predetermined α : if $p\text{-value} \leq \alpha$, reject H_0 ; otherwise fail to reject H_0 . State evidence for H_a in context; never report the p -value as $P(H_0 \text{ is true})$. Statistical significance does not by itself establish practical importance or causation. A random sample supports inference to the sampled population, while cause requires random assignment.

Finish after the video:

DOK 1 (a) Define the population proportion p , state H_0 and H_a , and calculate $\hat{p}$ and its signed difference from 0.60.

DOK 2 (b) Verify the Random, 10 percent, and Large Counts conditions. Calculate the one-proportion z statistic and lower-tail p -value.

DOK 3 (c) Adjudicate both statements and give the strongest conclusion at $\alpha = 0.05$. Interpret 0.0354 conditionally, distinguish significance from practical importance, bound cause and scope, and give the reader consequence of each overreach.

Frame: Because p -value = _____ is _____ α , I would _____ H_0 and conclude _____.

Frame: The result is statistically _____, but the design does not show _____; the 5.6-point gap's practical meaning depends on _____.

EXIT — turn this in

One thing the video changed about my first take: _____